

PHYSICAL AI AT WORK



At the ABB Bengaluru plant, the quality inspection process has been taken over by an AI-powered robot

● **At ABB Smart Factory, an AI-powered camera has taken over the quality check process for MCCBs. The algorithm is trained to look for defects such as unclear or incorrect printing, structural defects like cracks or uneven terminal thickness, and other anomalies. The company has had zero returns since the system was fully implemented.**

● **Tata Steel has implemented over 650 AI models across all its factories, utilising 50 billion AI tokens. Given the risk of accidents in factories and mines, the company uses soft sensors to gauge parameters like temperature and heat from an integrated maintenance facility. AI-driven fleet management and**



Tata Steel has set up a remote maintenance centre to monitor mining operations

predictive analytics monitor equipment health in real time, enabling maintenance before failure. This has raised asset availability and reduced unplanned downtime. A connected workforce platform, coupled with visual intelligence, helps detect unsafe conditions like PPE violations or unauthorised entry in hazardous zones.

● **Amazon Fresh uses computer vision models and Wi-fi and IoT enabled cameras to identify defects like cuts or bruises in fresh produce with shelf cameras automatically taking pictures at pre-determined intervals and assessing them for defects. This entire process takes six seconds.**

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A SLOW START

As with any new technology, some companies are resistant to adopting physical AI while some want to go all out and upgrade all the systems, sometimes without considering how effective it will be for them. If it's a set process, like tightening screws, a standard automation software is likely to do the job just as well, say experts. "If it is a critical process and adds value to the customer or improves the quality of the output, AI can make a difference," says ABB's Saju.

The technology is still at a nascent stage and quite expensive. At the same time, labour costs are comparatively low in India, so the decision will depend on what you want to achieve. "If you want the output from the

process line to not be dependent on who is working the line at the time, then there's a case to be made for intelligent automation," he says.

WAREHOUSE UPGRADES

Beyond the shop floor, physical AI solutions are also finding their way into warehouses and supply chains. "Technology plays a pivotal role throughout Amazon Fresh's supply chain. At the sourcing stage, we have built an ecosystem with a team of qualified agronomists offering expertise to farmer partners," says Srikant Sree Ram, Director, Amazon Fresh India. This improves farm yield and product quality through machine learning and computer vision-based algorithms